Power BI Assignment 2 – DAX & Data Visualization

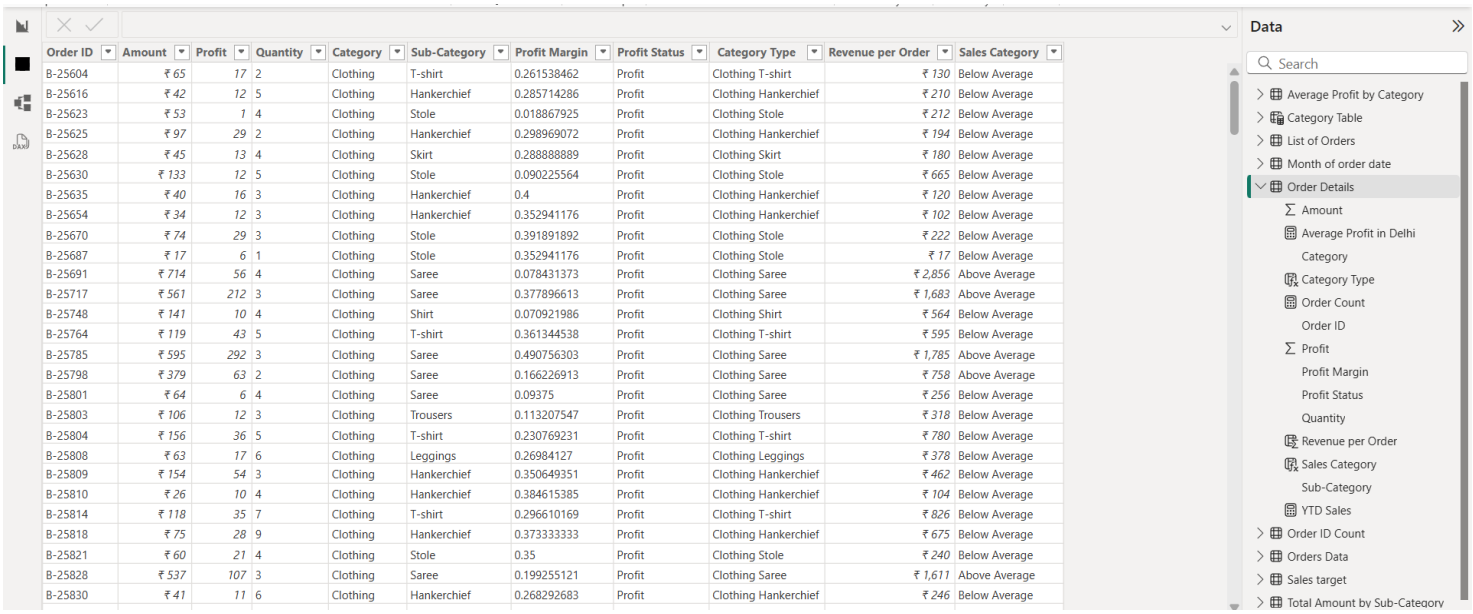
DAX Calculations

Calculated Columns

- **Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.**

- **Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.**

- **Create a Calculated Column to Categorize Sales: Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.**

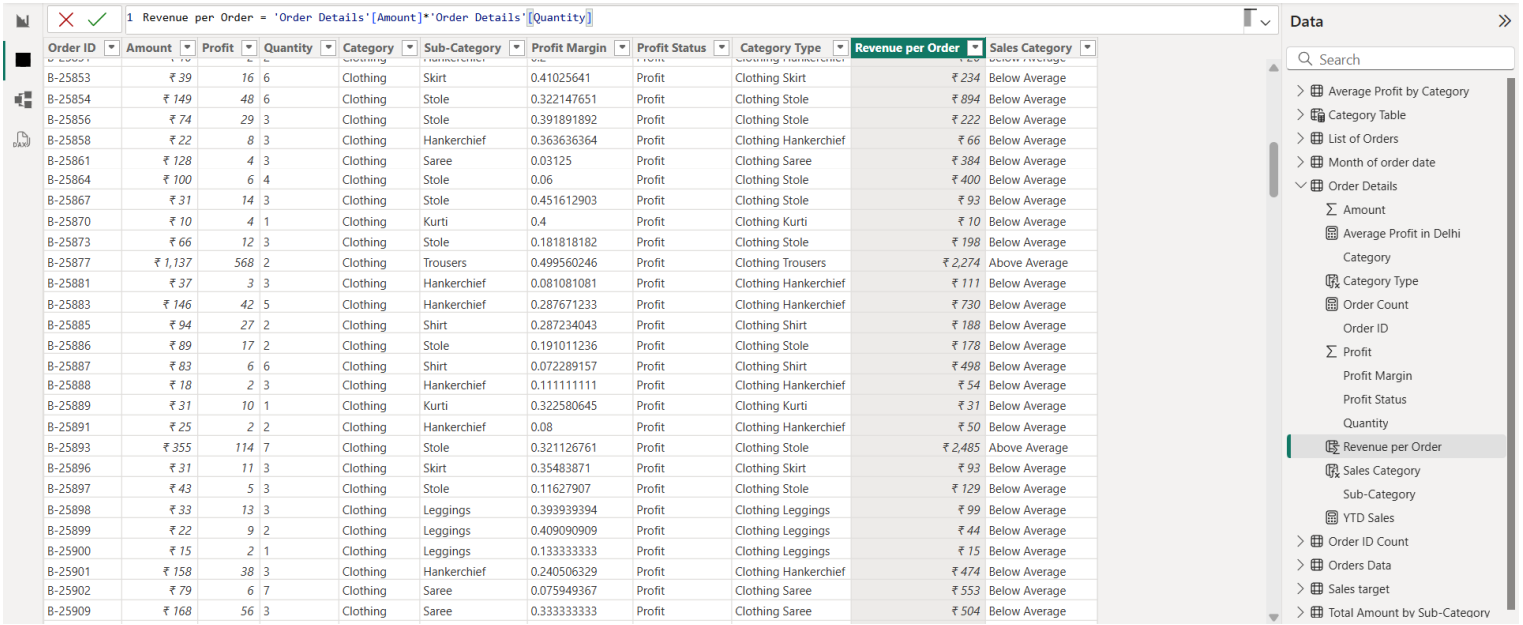


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
B-25604	₹ 65	17	2	Clothing	T-shirt	0.261538462	Profit	Clothing T-shirt	₹ 130	Below Average
B-25616	₹ 42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing Hankerchief	₹ 210	Below Average
B-25623	₹ 53	1	4	Clothing	Stole	0.018867925	Profit	Clothing Stole	₹ 212	Below Average
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing Hankerchief	₹ 194	Below Average
B-25628	₹ 45	13	4	Clothing	Skirt	0.288888889	Profit	Clothing Skirt	₹ 180	Below Average
B-25630	₹ 133	12	5	Clothing	Stole	0.090225564	Profit	Clothing Stole	₹ 665	Below Average
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing Hankerchief	₹ 120	Below Average
B-25654	₹ 34	12	3	Clothing	Hankerchief	0.352941176	Profit	Clothing Hankerchief	₹ 102	Below Average
B-25670	₹ 74	29	3	Clothing	Stole	0.391891892	Profit	Clothing Stole	₹ 222	Below Average
B-25687	₹ 17	6	1	Clothing	Stole	0.352941176	Profit	Clothing Stole	₹ 17	Below Average
B-25691	₹ 714	56	4	Clothing	Saree	0.078431373	Profit	Clothing Saree	₹ 2,856	Above Average
B-25717	₹ 561	212	3	Clothing	Saree	0.377896613	Profit	Clothing Saree	₹ 1,683	Above Average
B-25748	₹ 141	10	4	Clothing	Shirt	0.070921986	Profit	Clothing Shirt	₹ 564	Below Average
B-25764	₹ 119	43	5	Clothing	T-shirt	0.361344538	Profit	Clothing T-shirt	₹ 595	Below Average
B-25785	₹ 595	292	3	Clothing	Saree	0.490756303	Profit	Clothing Saree	₹ 1,785	Above Average
B-25798	₹ 379	63	2	Clothing	Saree	0.166226913	Profit	Clothing Saree	₹ 758	Above Average
B-25801	₹ 64	6	4	Clothing	Saree	0.09375	Profit	Clothing Saree	₹ 256	Below Average
B-25803	₹ 106	12	3	Clothing	Trousers	0.113207547	Profit	Clothing Trousers	₹ 318	Below Average
B-25804	₹ 156	36	5	Clothing	T-shirt	0.230769231	Profit	Clothing T-shirt	₹ 780	Below Average
B-25808	₹ 63	17	6	Clothing	Leggings	0.26984127	Profit	Clothing Leggings	₹ 378	Below Average
B-25809	₹ 154	54	3	Clothing	Hankerchief	0.350649351	Profit	Clothing Hankerchief	₹ 462	Below Average
B-25810	₹ 26	10	4	Clothing	Hankerchief	0.384615385	Profit	Clothing Hankerchief	₹ 104	Below Average
B-25814	₹ 118	35	7	Clothing	T-shirt	0.296610169	Profit	Clothing T-shirt	₹ 826	Below Average
B-25818	₹ 75	28	9	Clothing	Hankerchief	0.373333333	Profit	Clothing Hankerchief	₹ 675	Below Average
B-25821	₹ 60	21	4	Clothing	Stole	0.35	Profit	Clothing Stole	₹ 240	Below Average
B-25828	₹ 537	107	3	Clothing	Saree	0.199255121	Profit	Clothing Saree	₹ 1,611	Above Average
B-25830	₹ 41	11	6	Clothing	Hankerchief	0.268292683	Profit	Clothing Hankerchief	₹ 246	Below Average

Data

Search

- Average Profit by Category
- Category Table
- List of Orders
- Month of order date
- Order Details
  - Amount
  - Average Profit in Delhi
  - Category
  - Category Type
  - Order Count
  - Order ID
  - Profit
  - Profit Margin
  - Profit Status
  - Quantity
  - Revenue per Order
  - Sales Category
  - Sub-Category
  - YTD Sales
- Order ID Count
- Orders Data
- Sales target
- Total Amount by Sub-Category



```
1 Revenue per Order = 'Order Details'[Amount]*'Order Details'[Quantity]
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
B-25853	₹ 39	16	6	Clothing	Skirt	0.41025641	Profit	Clothing Skirt	₹ 234	Below Average
B-25854	₹ 149	48	6	Clothing	Stole	0.322147651	Profit	Clothing Stole	₹ 894	Below Average
B-25856	₹ 74	29	3	Clothing	Stole	0.391891892	Profit	Clothing Stole	₹ 222	Below Average
B-25858	₹ 22	8	3	Clothing	Hankerchief	0.363636364	Profit	Clothing Hankerchief	₹ 66	Below Average
B-25861	₹ 128	4	3	Clothing	Saree	0.03125	Profit	Clothing Saree	₹ 384	Below Average
B-25864	₹ 100	6	4	Clothing	Stole	0.06	Profit	Clothing Stole	₹ 400	Below Average
B-25867	₹ 31	14	3	Clothing	Stole	0.451612903	Profit	Clothing Stole	₹ 93	Below Average
B-25870	₹ 10	4	1	Clothing	Kurti	0.4	Profit	Clothing Kurti	₹ 10	Below Average
B-25873	₹ 66	12	3	Clothing	Stole	0.181818182	Profit	Clothing Stole	₹ 198	Below Average
B-25877	₹ 1,137	568	2	Clothing	Trousers	0.499560246	Profit	Clothing Trousers	₹ 2,274	Above Average
B-25881	₹ 37	3	3	Clothing	Hankerchief	0.081081081	Profit	Clothing Hankerchief	₹ 111	Below Average
B-25883	₹ 146	42	5	Clothing	Hankerchief	0.287671233	Profit	Clothing Hankerchief	₹ 730	Below Average
B-25885	₹ 94	27	2	Clothing	Shirt	0.287234043	Profit	Clothing Shirt	₹ 188	Below Average
B-25886	₹ 89	17	2	Clothing	Stole	0.191011236	Profit	Clothing Stole	₹ 178	Below Average
B-25887	₹ 83	6	6	Clothing	Shirt	0.072289157	Profit	Clothing Shirt	₹ 498	Below Average
B-25888	₹ 18	2	3	Clothing	Hankerchief	0.111111111	Profit	Clothing Hankerchief	₹ 54	Below Average
B-25889	₹ 31	10	1	Clothing	Kurti	0.322580645	Profit	Clothing Kurti	₹ 31	Below Average
B-25891	₹ 25	2	2	Clothing	Hankerchief	0.08	Profit	Clothing Hankerchief	₹ 50	Below Average
B-25893	₹ 355	114	7	Clothing	Stole	0.321126761	Profit	Clothing Stole	₹ 2,485	Above Average
B-25896	₹ 31	11	3	Clothing	Skirt	0.35483871	Profit	Clothing Skirt	₹ 93	Below Average
B-25897	₹ 43	5	3	Clothing	Stole	0.11627907	Profit	Clothing Stole	₹ 129	Below Average
B-25898	₹ 33	13	3	Clothing	Leggings	0.393939394	Profit	Clothing Leggings	₹ 99	Below Average
B-25899	₹ 22	9	2	Clothing	Leggings	0.409090909	Profit	Clothing Leggings	₹ 44	Below Average
B-25900	₹ 15	2	1	Clothing	Leggings	0.133333333	Profit	Clothing Leggings	₹ 15	Below Average
B-25901	₹ 158	38	3	Clothing	Hankerchief	0.240506329	Profit	Clothing Hankerchief	₹ 474	Below Average
B-25902	₹ 79	6	7	Clothing	Saree	0.075949367	Profit	Clothing Saree	₹ 553	Below Average
B-25909	₹ 168	56	3	Clothing	Saree	0.333333333	Profit	Clothing Saree	₹ 504	Below Average

Data

Search

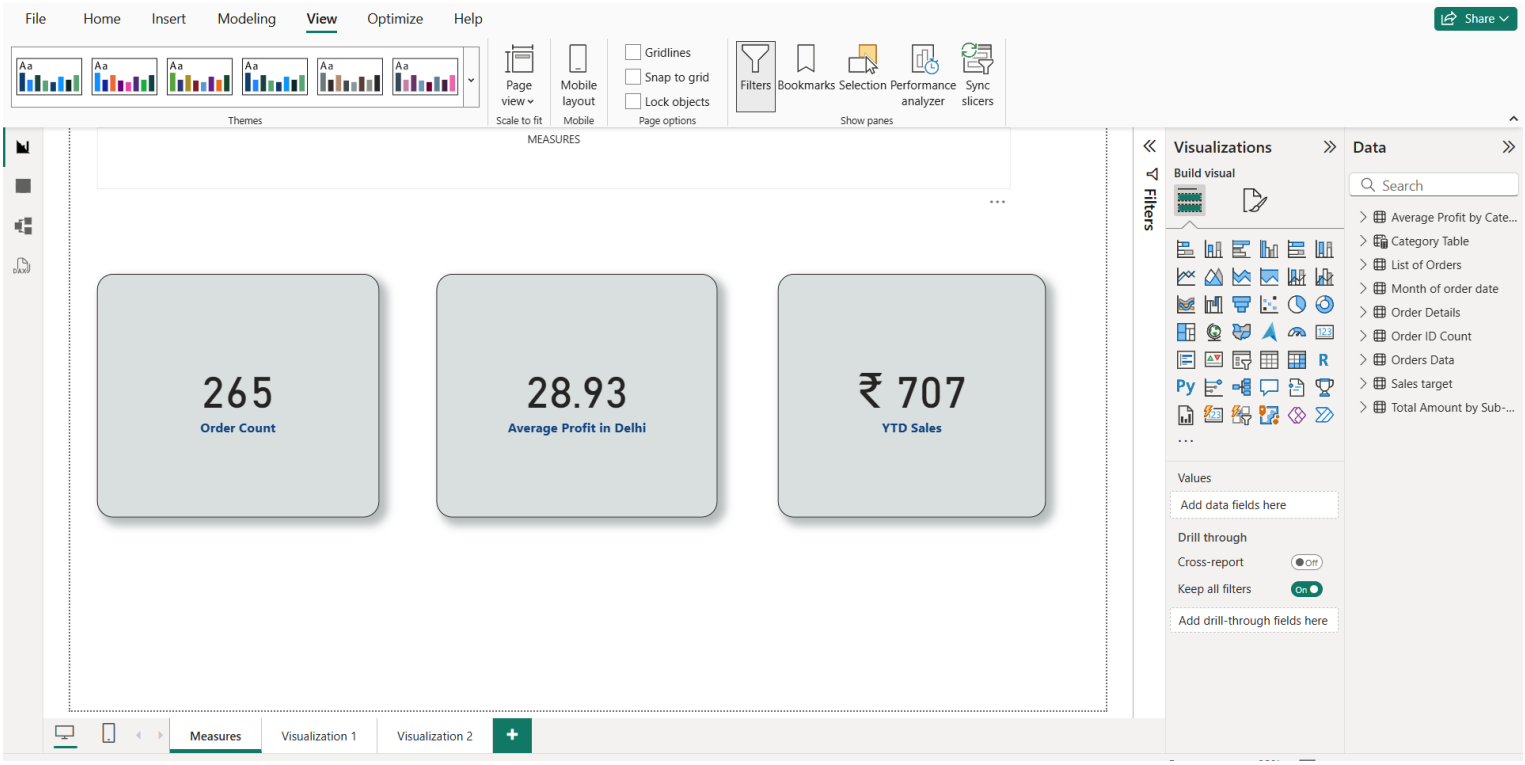
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  - Category
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  - Profit
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  - Profit Status
  - Quantity
  - Revenue per Order
  - Sales Category
  - Sub-Category
  - YTD Sales
- Order ID Count
- Orders Data
- Sales target
- Total Amount by Sub-Category

Calculated Measures

1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.

2. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.

3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.



Data Visualization

Create the following visualizations using Power BI:

● Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.

● Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.

● Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.

● Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

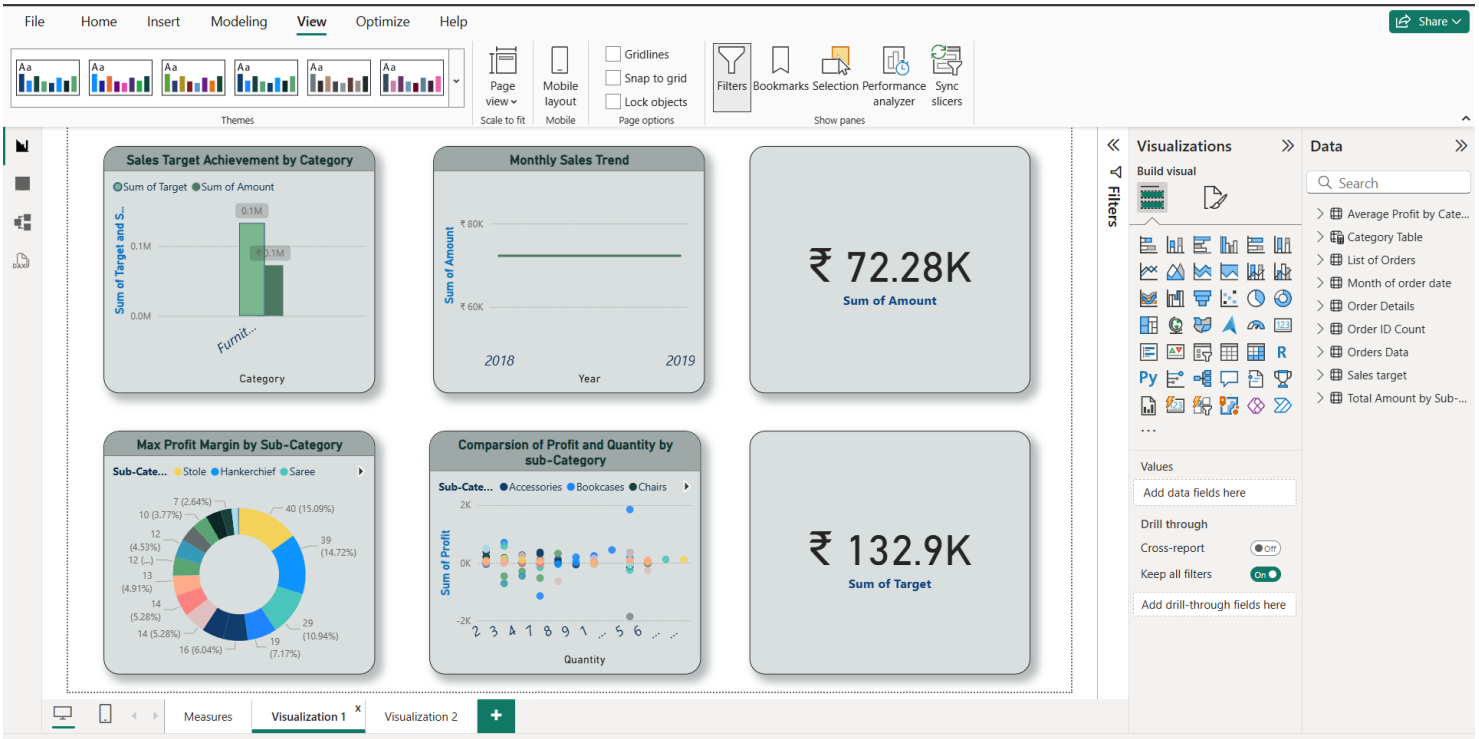
● **Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.**

● **Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.**

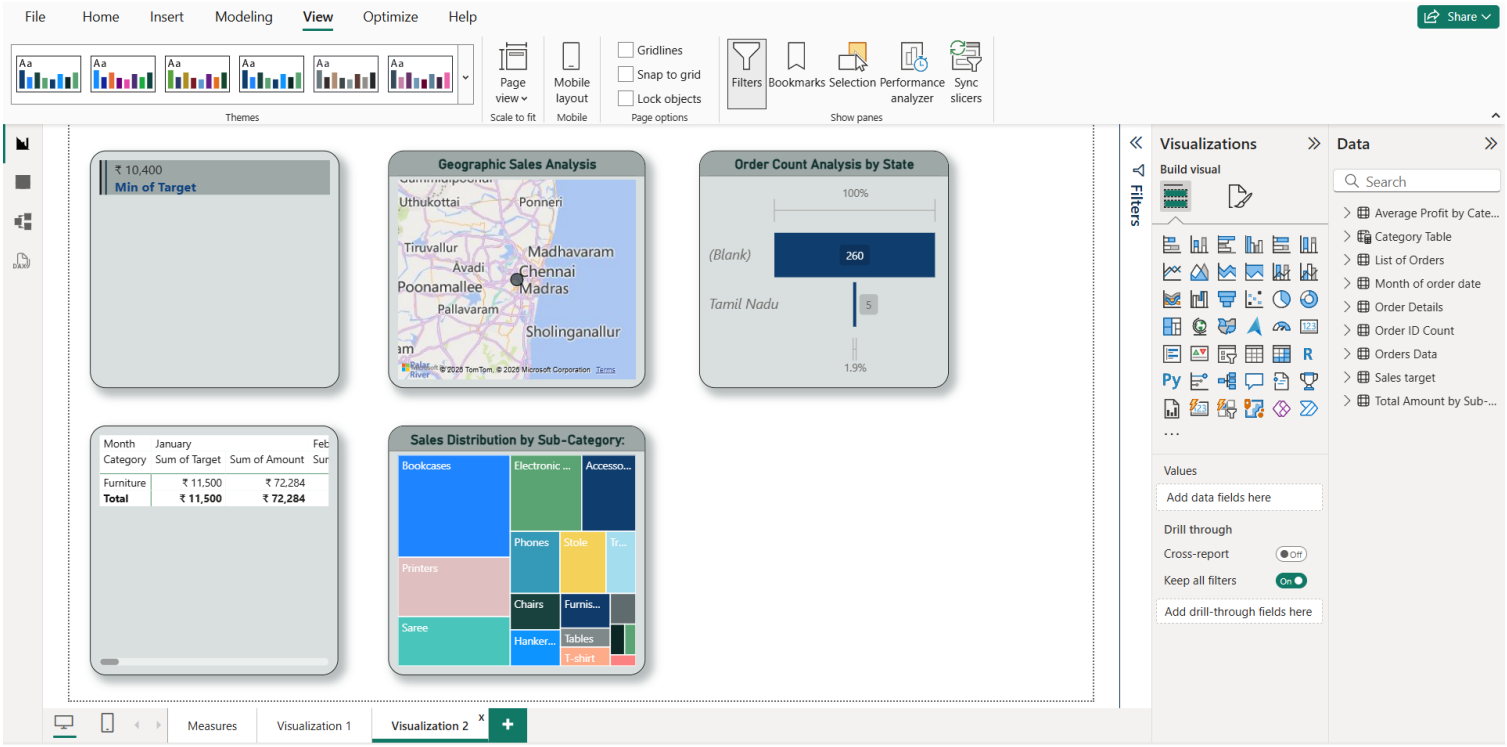
● **Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.**

● **Sales Distribution by Sub-Category: Represent the sales distribution across different**

● **sub-categories using a treemap.**



● **Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states**


File
Home
Insert
Modeling
View
Optimize
Help
Share
Themes
Page view
Scale to fit
Mobile layout
Mobile
Gridlines
Snap to grid
Lock objects
Page options
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Show panes
₹ 10,400
Min of Target
Geographic Sales Analysis
Uthukottai
Ponneri
Tiruvallur
Madhavaram
Avadi
Chennai
Poonamallee
Madras
Pallavaram
Sholinganallur
Order Count Analysis by State
100%
(Blank)
260
Tamil Nadu
5
1.9%
Month
Category
January
Sum of Target
Sum of Amount
Furniture
₹ 11,500
₹ 72,284
Total
₹ 11,500
₹ 72,284
Sales Distribution by Sub-Category:
Bookcases
Electronic ...
Accesso...
Phones
Stole
Printers
Chairs
Furnis...
Saree
Hanker...
Tables
T-shirt
Measures
Visualization 1
Visualization 2
Filters
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Data
Search
Average Profit by Cate...
Category Table
List of Orders
Month of order date
Order Details
Order ID Count
Orders Data
Sales target
Total Amount by Sub-...